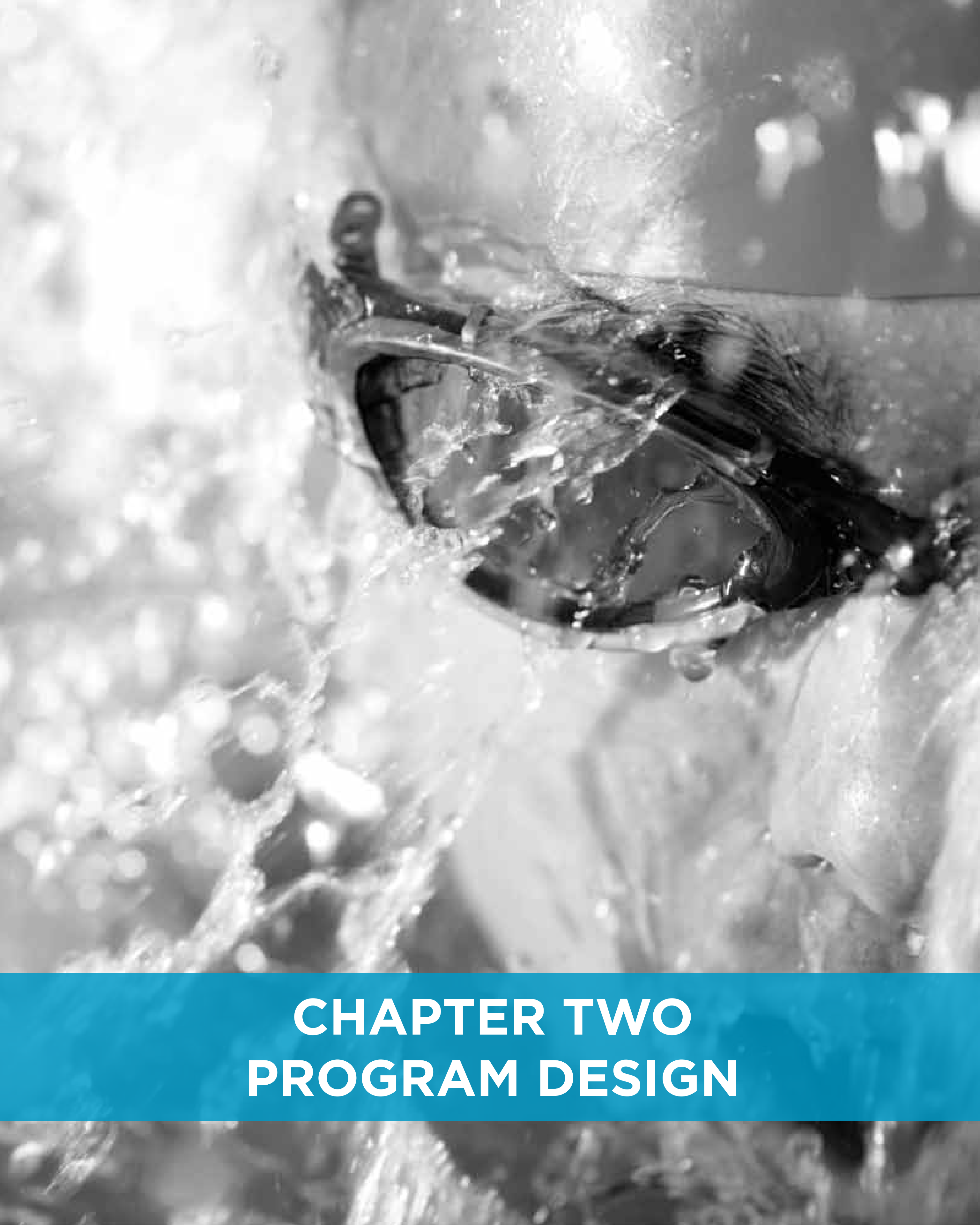

CHAPTER TWO
PROGRAM DESIGN

This chapter provides information that pertains to planning the design of strength training and conditioning programs. Program design is an overview perspective on what strength training and conditioning coaches should do to prepare athletes for their preparatory and competitive seasons. Properly designed strength training and conditioning programs are vital to any athletic preparation. Strength training and conditioning programs can improve athletic performance, decrease the likelihood of injuries, and allow for the development of sound weightlifting and basic resistance training techniques (18,22,23,30,35). Development of these attributes allows athletes to gain a strong foundation of fundamental strength training and conditioning practices, and be adequately prepared for advanced strength training and conditioning programs (e.g., collegiate training programs).

How Do We Organize Training?

Training theory uses models of training application that have evolved over 60 years. A model is a simplification or simulation of a real-world complex process (8,9,10). One should know that a model is never exactly like the real-world situation, but attempts to simulate the most important aspects that need attention while ignoring those factors that are considered unimportant. The term used to describe the special planning that occurs with athletic training is "periodization." Periodization is composed of two major concepts that occur simultaneously. Training is divided into "periods," and these periods are cyclic in that various aspects of training are repeated and form a system (31).

Training Design Terminology

As with every other area covered in this manual, we have had to cover new terms that are a part of the knowledge area, and sometimes the culture of a particular facet of strength training and conditioning. The following list is a summary of some important terms that will be needed to carry on further in the study of training design.

Specific Adaptations to Imposed Demands (SAID Principle)

The SAID principle is a fundamental principle in the field of strength training and conditioning (7). The SAID principle states that an athlete's body will adapt to exactly what is demanded of it—no more and no less. This principle says that you must give the athlete's body an unambiguous message of what you want it to become by providing training stressors that mimic all, or parts, of the target physical capacities or skills. The SAID principle constrains strength training and conditioning coaches in their program designs to achieve specific adaptations based on the demands put on the system.

Annual Plan

Training theory goes to considerable lengths to describe the time dimension in designing training tasks. An annual plan is the calendar-based approach used to place the various demands of training within a calendar year. Usually, the annual plan begins immediately following the last competition of the previous season, and ends after the last competition of the succeeding season.

Macrocycle

The terms "macrocycle" and "annual plan" have occasionally been used synonymously (17). The macrocycle, for our purposes, is the linking of the general physical preparation phase, the specific preparation phase, the pre-competitive phase, the competitive phase, and the peak phase. As such, often it is convenient to link together all of the work leading to a single championship contest. If there is more than one major competition per year, then the year may need more than one macrocycle, typically one for each major season. For example, spring ball and fall ball in football, and the indoor and outdoor seasons of track and field will need to be considered two macrocycles per annual plan (or year).

Mesocycle

A mesocycle is an intermediate duration of time planning that usually lasts from weeks to a few months. The mesocycle is perhaps the first functional unit for training planning where specific training goals may be assigned and achieved during a particular mesocycle. For example, a pre-competitive mesocycle for basketball might be a period of a month or so where the dominant form of training is scrimmaging. Or, a mesocycle may be assigned to a shot putter in the early season to use strength training and conditioning to enhance his/her maximal strength.

Microcycle

A microcycle is a smaller time division that lasts from one to a few weeks. The microcycle is the smallest unit of planning in which we can expect to see the beginnings of relatively stable adaptations. Typically, it takes about a week of consistent training demands for the body to be pushed or forced to adapt, and thus change its chemistry, biomechanics, and/or skills to adapt to the training demands.

Training Lesson

A training lesson is a single bout of training where the athlete begins a session with a warm-up, practices some aspect of the sport or strength training and conditioning, and then ends the session with a cool-down. A single training lesson is relatively powerless in influencing the adaptation of an athlete. Only by the accumulation of about a week's (or microcycle's) worth of training stimuli is the athlete forced to adapt to the new training demands.

Program

In the strength training and conditioning world, a program is the actual exercises, sets, repetitions, resistances, inter-set rest periods, inter-lesson rest periods, and so forth. An example program is provided at the end of this chapter.

Basis of Program Design Decisions

You will find nearly all program design literature to be triphasic (3 phases), or have three periods/stages. For example, nearly all program designs are based on a simple yet profound idea proffered by Hans Selye called the General Adaptation Syndrome (GAS) (49,50).

The GAS consists of three phases: alarm, resistance, and exhaustion (16,17). The athlete begins with a level of being called "homeostasis" (7). The alarm phase occurs when the athlete is presented with a large enough stressor to evoke fatigue (16,17). Stress is defined as anything that causes an organism, or in this case an athlete, to react (49,50). The alarm stage is distinguished by markers of fatigue, reduced performance abilities, and decreased physical capacities. The resistance phase occurs when the body temporarily adapts to the applied stressor and is able to cope physically with the demands. The resistance phase indicates that the athlete has achieved a level of adaptation that is actually greater or better than his/her previous homeostatic level. Finally, if the stressor is too great to continue to resist, the stimulation increases, or the athlete is not allowed to rest, then the body slips into the exhaustion phase. During the exhaustion phase, symptoms of the alarm phase return but the magnitude is greater and the fatigue much more profound. Figure 2-1 shows the three stages of Selye's stress adaptation model (overreaching, supercompensation, overtraining), and demonstrates how performance is affected in each of the three phases.

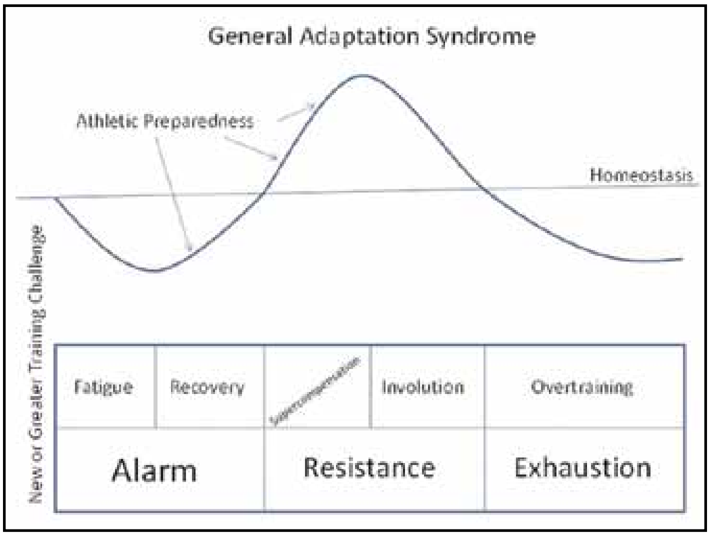

Figure 2-1. Selye's General Adaptation Syndrome (49,50)

In Figure 2-1, athlete preparedness is represented by a curved line at the top of the graphic. Homeostasis is the horizontal line labeled at the right. Note that the line that represents athletic preparedness shows that the athlete fatigues, recovers, and then supercompensates (enters the resistance stage). If nothing else is done, the athlete returns to homeostasis via involution, or descends into overtraining and exhaustion due to the return of fatigue and the inability of the athlete to continue to compensate for the applied stressors. Figure 2-2 shows the Selye-type curves of adaptation together with an example of training loads and how the training loads, adaptation, rest periods, and timing work together.

Training Load Prescriptions

Training theory is not useful unless it can be applied to the real world of training. Figure 2-2 shows how training loads are applied in a systematic fashion to take advantage of the GAS, and systematically improve athletic performance. Note that the principle of progression is important when prescribing training loads throughout a program to ensure proper, and desired, adaptations.

Rules for Exercise Selection and Prescription

The field of strength training and conditioning is driven by research-based information that allows for the proper utilization of sets, repetitions, volume, and rest periods to elicit desired physiological adaptations. The information surrounding these topics has been a topic of discussion for many years. Professional organizations have guidelines and recommendations associated with these variables that allow strength training and conditioning coaches to target specific adaptations to the physiological system (based on the guidelines provided) (3,4,5,24,26). It is imperative that strength training and conditioning coaches understand these guidelines, and the application of these guidelines, to induce the desired adaptations for the athletes that they work with. These guidelines and recommendations not only promote certain training adaptations, but also decrease the likelihood of injuries during training. Direct access to these guidelines and recommendations should be available to strength training and conditioning coaches at any facility. Access to this information will not only decrease the chance of injuries (due to improper programming) but also provide correct prescription to target physiological variables progressively during the course of a training macrocycle. Refer to the sample program at the end of this chapter for further information on program design.

Figure 2-3 shows the spectrum of various repetition ranges and their relative placement over time. Note that repetition ranges will be dictated by the resistance involved, such that one should attempt to keep the resistance in line with the repetitions diagramed in Figure 2-3.

Warm-Up and Stretching

A warm-up is designed to prepare an athlete for training or competition, and can improve subsequent performance and lessen the risk of injuries. A warm-up should be incorporated in every program design. A warm-up period is important before any athletic performance or physical activity, the goal being to prepare the athlete mentally and physically for exercise or competition. A well-designed warm-up should increase muscle temperature, circulation, and provide an opportunity for skill rehearsal (1,28,37,57). Moreover, one of the aspects of a warm-up is simply to "stir" cellular content so the sarcoplasm becomes liquefied (43,45,47,58). These warm-up effects can have the following positive impacts on performance:

- Faster muscle contraction and relaxation of both agonist and antagonist muscles (33)

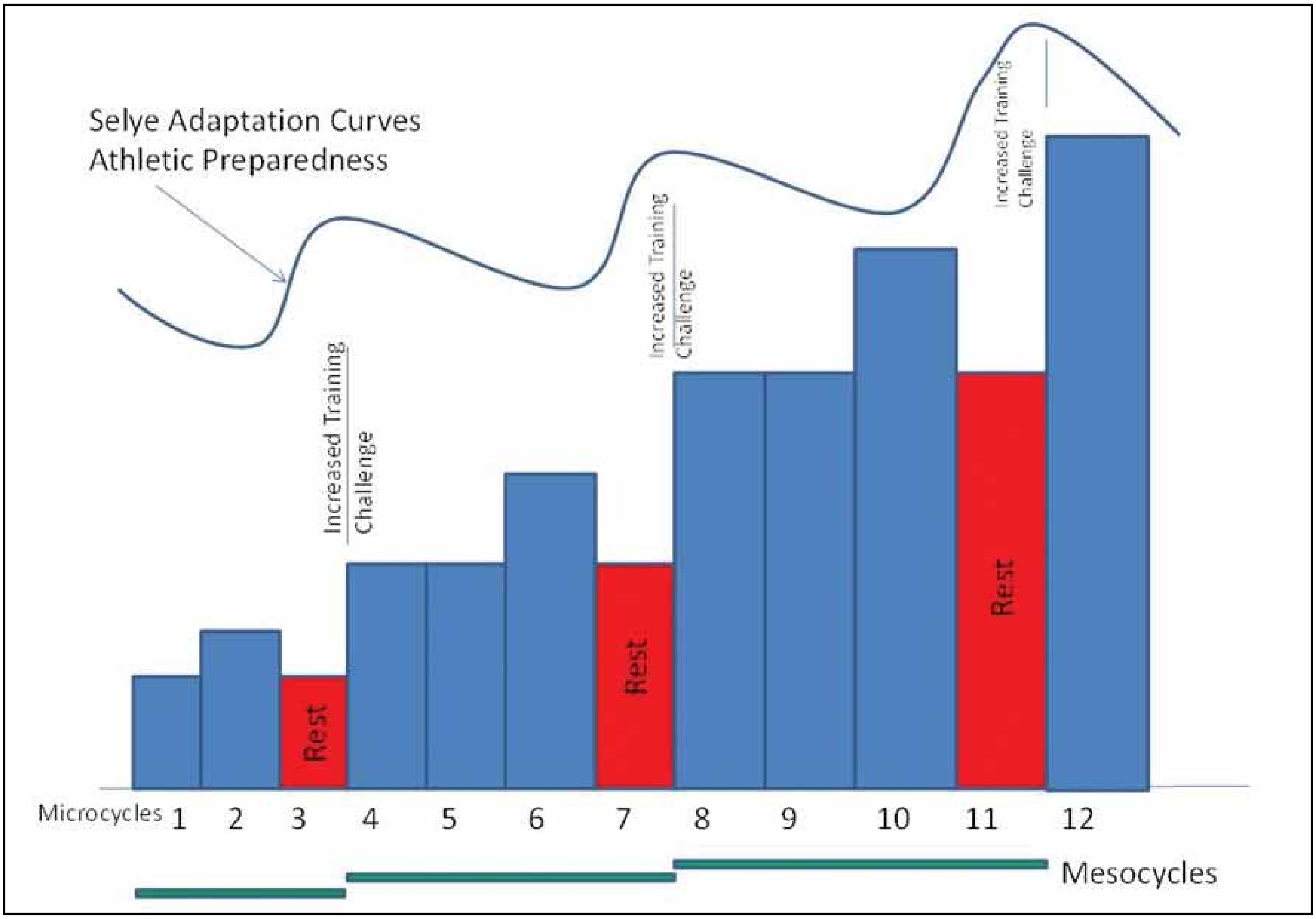

Figure 2-2. Combination of Training Design Aspects in Relation to the General Adaptation Syndrome (49,50)

- Improvements in the rate of force development and reaction time (2)
- Improvements in muscle strength and power (14,24)
- Lowered viscous resistance in muscles (6,42,47,58)
- Improved oxygen delivery due to the Bohr effect, whereby higher temperatures facilitate
- oxygen release from hemoglobin and myoglobin (39)
- Increased blood flow to active muscles (39)
- Enhanced metabolic reactions (24)

While the influence of a warm-up on injury prevention is unclear, the evidence suggests a positive effect, or no effect at all, on injury (15,21,29,52,53,59). The relationship between stretching and injury prevention is tenuous at best.

Components of a Warm-Up

A total warm-up program includes the following two components (17,28,31,37):

A general warm-up period may consist of 5 – 10 min of slow activity, such as jogging or skipping. Alternatively, low-intensity sport-specific actions, such as dribbling a soccer ball, can be productive during this time. This provides a very sport-specific general warm-up that aids in skill development and raises body temperature. The aim of this period is to increase heart rate, blood flow, deep muscle temperature, respiration rate, perspiration, and decrease viscosity of joint fluids.

A specific warm-up period incorporates movements similar to the movements of the athlete's sport. It involves 8 – 12 min of dynamic stretching that focuses on movements that work through the range of motion required for the sport, such as the walking knee lift. Sport-specific movements of increasing intensity, such as

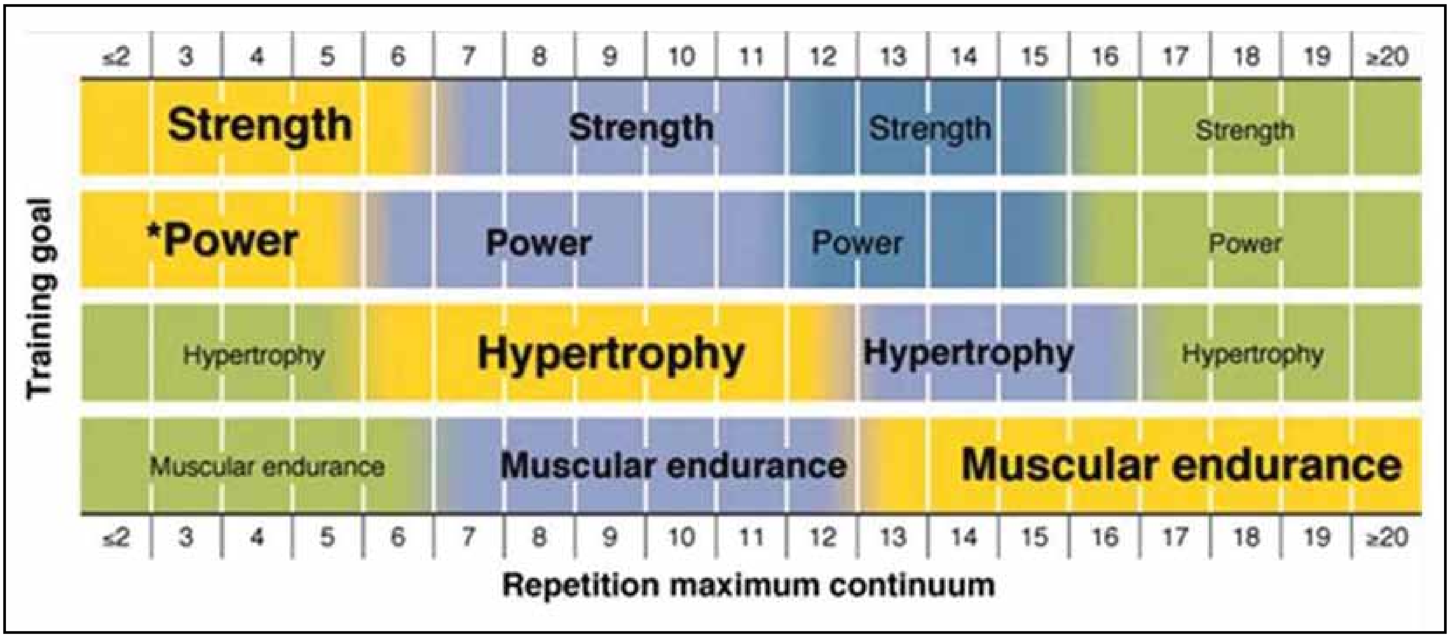

Figure 2-3. Repetition Ranges for Specific Training Outcomes

sprint drills, bounding activities, or jumping, follow the dynamic stretching. The more power necessary for the sport or activity, the more important the warm-up becomes. Including high-intensity dynamic exercises can facilitate subsequent performance. This phase should also include rehearsal of the skills to be performed.

A warm-up should progress gradually and provide sufficient intensity to increase muscle and core temperatures without causing fatigue or reducing energy stores. It is likely that there are optimal levels of warm-ups related to the specific sport, the athlete, and the environment, so no one warm-up routine is best for every athlete, or sport.

Stretching During Warm-Up

There are four main types of stretching: static, ballistic, dynamic, and proprioceptive neuromuscular facilitation (PNF). Static stretching has long been used in a warm-up, with the aim of enhancing performance. However, recent reviews of the literature surrounding the role of static stretching question this practice (51,52). There is little, if any, evidence that stretching pre- or post-participation prevents injury or subsequent muscle soreness (32,34,46,48,51,52). Although static stretching before activity may increase performance in sports that require an increased range of motion, such as gymnastics, stretching activities may or may not influence subsequent strength, power, running speed, reaction time, and strength endurance performance, and often depends on what intermittent activity was used between the stretching and the strength activity (11,12,13,19,20,36,40,41,44,48,54,55,56). In these cases it is important that the strength training and conditioning professional performs a benefit–risk analysis when choosing whether or not to include static stretching in a warm-up (38).

Dynamic stretching, which is functionally based and sues sport-specific movements to prepare for activity, does not seem to elicit the performance reduction effects of static and PNF stretching, but has been shown to improve subsequent running performance (27,60,61). Given these findings, the use of static, PNF, and ballistic stretching in a warm-up needs to be questioned. Based on current evidence, dynamic stretching is the preferred option for stretching during a warm-up.

The degree of stretching required in a warm-up depends on the type of sport. Sports that require increased flexibility, such as gymnastics or diving, require a greater degree of stretching (52). Additionally, those sports with high demands for a stretch-shortening cycle of high-intensity, as in sprinting and American football, are likely to require more stretching than those with low or medium stretch-shortening cycle activity, as in jogging or cycling (59). Strength training and conditioning professionals should look at the specific range of motion and stretch-shortening cycle requirements of the sport or activity and use this information to design an appropriate warm-up routine.

Conclusion

Program design must take into account many factors related to the life and training of each athlete. Program design is guided by theory and hands-on experience from many previous and current coaches and sport scientists. The following provides a 12-week strength and conditioning program as an example of how to develop and implement the lessons presented throughout this manual.

Sample Strength and Conditioning 12-Week Program

Base Phase

Basics of Strength and Conditioning - Twelve Week Program												
Monday - Explosive	**Week 1 - 65%**			**Week 2 - 70%**			**Week 3 - 65%**			**Week 4 - 75%**		
	Load			*Load*			*Unload*			*Load*		
Lifting Warm-up												
BB Rack Shrug OR Rack Jump OR Rack Clean	2x5			3x5			3x5			3x5		
BB High Pull	2x5			3x5			3x5			3x5		
BB Standing Shoulder Press OR DB Shoulder Raises	2x5			3x5			3x5			3x5		
Pulling Choice	2x10			3x10			3x10			3x10		
Bicep Choice	2x10			3x10			3x10			3x10		
Ab Planks	2x 15-30 seconds			3x 15-30 seconds			3x 30-45 seconds			3x 30-45 seconds		
Tuesday - Strength	**Week 1 - 60%**			**Week 2 - 65%**			**Week 3 - 60%**			**Week 4 - 70%**		
Lifting Warm-up												
BB Front Squat OR Modified Squat	2x10			3x10			3x10			3x10		
BB Romanian Deadlift (RDL)	2x10			3x10			3x10			3x10		
Single Leg Choice	2x10ea			3x10ea			3x10ea			3x10ea		
BB Incline Bench Press	2x10			3x10			3x10			3x10		
Triceps Choice	2x10			3x10			3x10			3x10		
AB Choice	2x10			3x10			3x10			3x10		
Thursday - Explosive	**Week 1 - 60%**			**Week 2 - 65%**			**Week 3 - 60%**			**Week 4 - 70%**		
Lifting Warm-up												
BB Rack Shrug OR Rack Jump OR Rack Clean	2x5			3x5			3x5			3x5		
BB High Pull	2x5			3x5			3x5			3x5		
BB Standing Shoulder Press OR DB Shoulder Raises	2x5			3x5			3x5			3x5		
Pulling Choice	2x10			3x10			3x10			3x10		
Bicep Choice	2x10			3x10			3x10			3x10		
Ab Planks	2x 15-30 seconds			3x 15-30 seconds			3x 30-45 seconds			3x 30-45 seconds		
Friday - Strength	**Week 1 - 65%**			**Week 2 - 70%**			**Week 3 - 65%**			**Week 4 - 75%**		
Lifting Warm-up												
BB Back Squat OR Modified Squat	2x10			3x10			3x10			3x10		
BB Romanian Deadlift (RDL)	2x10			3x10			3x10			3x10		
Single Leg Choice	2x10ea			3x10ea			3x10ea			3x10ea		
BB Bench Press	2x10			3x10			3x10			3x10		
Triceps Choice	2x10			3x10			3x10			3x10		
AB Choice	2x10			3x10			3x10			3x10		
Key	**Comments/Notes**											
BB - Barbell / DB - Dumbbell												
CB - Cable / MB - Medicine Ball												

Strength Phase

Basics of Strength and Conditioning - Twelve Week Program				
Monday - Explosive	**Week 5 - 75%**	**Week 6 - 80%**	**Week 7 - 75%**	**Week 8 - 85%**
	Load	*Load*	*Unload*	*Load*
Lifting Warm-up				
BB Hang Shrug OR Hang Jump OR Hang Clean	3x5	3x5	3x5	3x5
BB Push Press OR Standing Shoulder Press	3x5	3x5	3x5	3x5
Pulling Choice	3x5	3x5	3x5	3x5
Pulling Choice	3x5	3x5	3x5	3x5
Ab Planks	2x 15-30 seconds	3x 15-30 seconds	3x 30-45 seconds	3x 30-45 seconds
Tuesday - Strength	**Week 5 - 70%**	**Week 6 - 75%**	**Week 7 - 70%**	**Week 8 - 80%**
Lifting Warm-up				
BB Deadlift OR Modified Deadlift	3x5	3x5	3x5	3x5
BB Romanian Deadlift (RDL)	3x5	3x5	3x5	3x5
Single Leg Choice	3x5ea	3x5ea	3x5ea	3x5ea
BB Incline Bench Press OR DB Incline Bench Press	3x5	3x5	3x5	3x5
Triceps Choice	3x5	3x5	3x5	3x5
Thursday - Explosive	**Week 5 - 70%**	**Week 6 - 75%**	**Week 7 - 70%**	**Week 8 - 80%**
Lifting Warm-up				
BB Hang Shrug OR Hang Jump OR Hang Clean	3x5	3x5	3x5	3x5
BB Push Press OR Standing Shoulder Press	3x5	3x5	3x5	3x5
Pulling Choice	3x5	3x5	3x5	3x5
Pulling Choice	3x5	3x5	3x5	3x5
Ab Planks	2x 15-30 seconds	3x 15-30 seconds	3x 30-45 seconds	3x 30-45 seconds
Friday - Strength	**Week 5 - 75%**	**Week 6 - 80%**	**Week 7 - 75%**	**Week 8 - 85%**
Lifting Warm-up				
BB Back Squat OR Modified Squat	3x5	3x5	3x5	3x5
BB Romanian Deadlift (RDL)	3x5	3x5	3x5	3x5
Single Leg Choice	3x5ea	3x5ea	3x5ea	3x5ea
BB Bench Press OR DB Bench Press	3x5	3x5	3x5	3x5
Triceps Choice	3x5	3x5	3x5	3x5
Key	**Comments/Notes**			
BB - Barbell / DB - Dumbbell				
CB - Cable / MB - Medicine Ball				

Peak Phase

Basics of Strength and Conditioning - Twelve Week Program

Monday - Explosive	Week 9 - 75%			Week 10 - 80%			Week 11 - 75%			Week 12 - 85%		
	Load			*Load*			*Unload*			*Load*		
Lifting Warm-up												
BB Clean Shrug OR Clean Jump OR Clean	3x3			3x3			3x3			3x3		
BB Push Jerk OR Push Press	3x3			3x3			3x3			3x3		
Pulling Choice	3x3			3x3			3x3			3x3		
Ab Planks	2x 15-30 seconds			3x 15-30 seconds			3x 30-45 seconds			3x 30-45 seconds		
Tuesday - Strength	**Week 9 - 70%**			**Week 10 - 75%**			**Week 11 - 70%**			**Week 12 - 80%**		
Lifting Warm-up												
BB Front Squat OR Deadlift OR Modified	3x3			3x3			3x3			3x3		
BB Romainian Deadlift (RDL)	3x3			3x3			3x3			3x3		
Single Leg Choice	3x3ea			3x3ea			3x3ea			3x3ea		
BB Incline Bench Press OR DB Incline Bench Press	3x5			3x5			3x5			3x5		
Thursday - Explosive	**Week 9 - 70%**			**Week 10 - 75%**			**Week 11 - 70%**			**Week12 - 80%**		
Lifting Warm-up												
BB Clean Shrug OR Clean Jump OR Clean	3x3			3x3			3x3			3x3		
BB Push Jerk OR Push Press	3x3			3x3			3x3			3x3		
Pulling Choice	3x3			3x3			3x3			3x3		
Ab Planks	2x 15-30 seconds			3x 15-30 seconds			3x 30-45 seconds			3x 30-45 seconds		
Friday - Strength	**Week 9 - 75%**			**Week 10 - 80%**			**Week 11 - 75%**			**Week 12 - 85%**		
Lifting Warm-up												
BB Back Squat OR Modified Squat	3x3			3x3			3x3			3x3		
BB Romainian Deadlift (RDL)	3x3			3x3			3x3			3x3		
Single Leg Choice	3x3ea			3x3ea			3x3ea			3x3ea		
BB Bench Press OR DB Bench Press	3x3			3x3			3x3			3x3		
Key	**Comments/Notes**											
BB - Barbell / DB - Dumbbell												
CB - Cable / MB - Medicine Ball												

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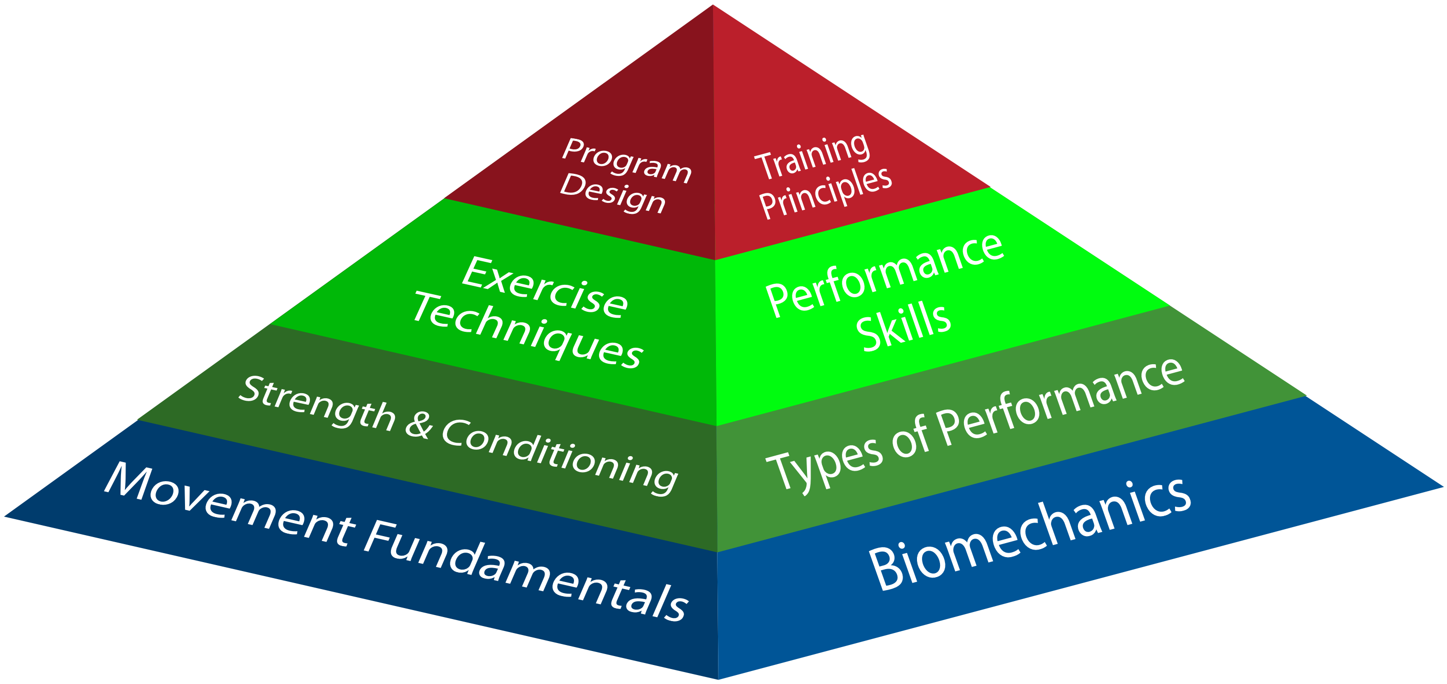

Performance Pyramid